



GPS Tracker Hardware Integration Specification

1. Connection Protocol

- **Protocol:** Raw TCP Socket
- **Target Port:** 5000
- **Target IP:** 192.168.1.213

2. Payload Format The tracker must send data as a **raw JSON string** over the TCP connection.

3. Required JSON Structure Every packet sent to the server **must** contain the following fields:

Field	Type	Description	Example
device_uid	String	The unique hardware identifier (IMEI/ID).	"8654020409"
name	String	The registered alias/name of the tracker.	"Delivery Van 1"
lat	Float/Number	Latitude coordinates.	13.0827
lon	Float/Number	Longitude coordinates.	80.2707
speed	Float/Number	Current speed of the vehicle in km/h.	45.5
battery	Integer	Current battery percentage (0-100).	85

4. Example Payload

```
{
  "device_uid": "8654020409",
  "name": "Delivery Van 1",
  "lat": 13.0827,
  "lon": 80.2707,
  "speed": 45.5,
  "battery": 85
}
```

5. Security & Validation Rules (CRITICAL) The backend employs a strict security validation process. When the server receives a packet, it checks the database for an active device matching **BOTH** the `device_uid` AND the `name`.

- If both match exactly: The data is accepted and logged.
- If there is a mismatch (e.g., the UID is correct but the name is wrong): The server will **reject and drop the payload completely** to prevent unauthorized data spoofing.

6. Server Acknowledgment Upon successfully validating and saving the JSON payload, the server will reply to the tracker over the TCP socket with an acknowledgment string in the following format: `ACK, [device_uid]#\n` (Example: `ACK,8654020409#\n`)